TRAVERSE ADJUSTMENT REPORT

SurveyX Digital
Date: 2026-10-23

PROJECT	**Al Khor Expressway**	LOCATION	**Al Khor, Qatar**
SURVEYOR	**Asjad Mehmood**	CHECKED BY	**Senior Surveyor**
INSTRUMENT	**Leica TS16 2"**	DATE	**2026-10-23**

1. Traverse details

Traverse type	**Closed (checked on a known point)**
Observations	**Site coordinates as measured**
Adjustment method	**Bowditch (Compass rule)**
Coordinate system	**Qatar National Grid (QND95), WGS84 latitude and longitude**
Station point	**CP-01 E 228450.312 N 441203.845**
Backsight point	**CP-02 E 228912.640 N 441655.118**
Reference bearing	**CP-01 -> CP-02 045°41'36" (646.061 m)**
Closing point	**CP-03 E 229265.871 N 441392.006**
Traverse legs	**5 (total length 931.579 m)**

2. Closure check

ACCEPTED - within the required precision

Misclosure dE = 12.0 mm dN = -9.0 mm Linear = 15.0 mm
Precision achieved 1 : 62,105 Required 1 : 10,000 Total length 931.579 m
Misclosure = measured closing coordinates - known closing coordinates. Corrections are the opposite sign.

3. Input coordinates (as measured on site)

Point	Easting (m)	Northing (m)	Remark
CP-01	228450.312	441203.845	Station (known)
TP-01	228610.458	441118.300	Steel pin
TP-02	228795.123	441090.770	Steel pin
TP-03	228968.548	441162.410	Steel pin
TP-04	229120.243	441281.652	Steel pin
CP-03	229265.883	441391.997	Closing point as measured

4. Output coordinates (adjusted)

Point	Corr. dE (mm)	Corr. dN (mm)	Adjusted Easting (m)	Adjusted Northing (m)
CP-01	0.0	0.0	**228450.312**	**441203.845**
TP-01	-2.3	1.8	**228610.456**	**441118.302**
TP-02	-4.7	3.6	**228795.118**	**441090.774**
TP-03	-7.2	5.4	**228968.541**	**441162.415**
TP-04	-9.6	7.2	**229120.233**	**441281.659**
CP-03	-12.0	9.0	**229265.871**	**441392.006**

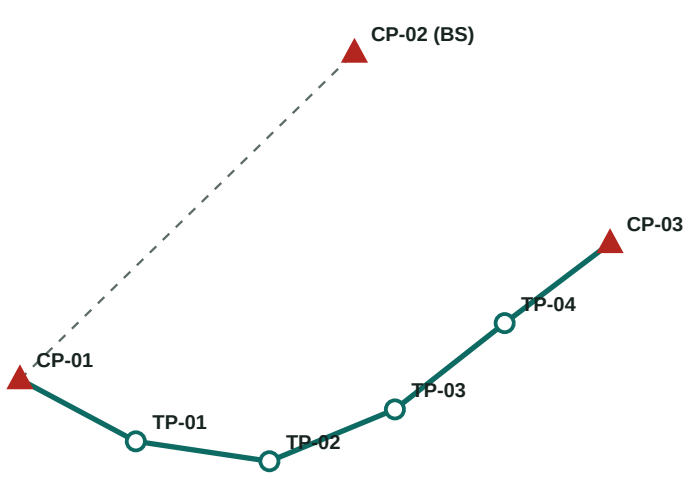
5. Bearings and distances (from output coordinates)

From	To	Bearing (WCB)	Distance (m)
CP-01	TP-01	118°06'35"	181.559
TP-01	TP-02	098°28'44"	186.703
TP-02	TP-03	067°33'15"	187.638
TP-03	TP-04	051°49'46"	192.950
TP-04	CP-03	052°50'58"	182.720

6. Latitude and longitude (WGS84, from the Qatar National Grid)

Point	Easting (m)	Northing (m)	Latitude	Longitude
CP-01	228450.312	441203.845	25.72512657	51.49940387
TP-01	228610.456	441118.302	25.72435130	51.50099789
TP-02	228795.118	441090.774	25.72409921	51.50283748
TP-03	228968.541	441162.415	25.72474249	51.50456721
TP-04	229120.233	441281.659	25.72581587	51.50608146
CP-03	229265.871	441392.006	25.72680903	51.50753519

7. Sketch



Sketch, not to a printed scale. Red triangles: known points. Circles: traverse points (adjusted).

SURVEYED BY

Asjad Mehmood

Signature / Date

CHECKED BY

Senior Surveyor

Signature / Date

APPROVED BY

Signature / Date